VMRT Seminar: Hilbert Schemes

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Lecture 1

The first step in studying families of rational curves is to construct the spaces which parameterize them. This lecture introduces Hilbert schemes through their universal property, explains the role of flatness and Hilbert polynomials, and constructs morphism schemes as open subschemes of Hilbert schemes by parameterizing graphs.

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Chapter 1

Hilbert Schemes and Morphism Schemes

In this lecture, k is a field and S is a noetherian scheme. Geometric examples may be read over $k = \mathbb{C}$. Schemes used as test objects are allowed to be nonreduced. For a scheme X over S and an S-scheme T, write

$$X_T = X \times_S T.$$

When discussing coherent sheaves and their cohomology, the bases are noetherian unless otherwise specified. In the moduli functors, families over arbitrary test schemes are required to be of finite presentation. This agrees with the usual coherent formulation over noetherian bases.

There are two parameter spaces to keep in view. A Hilbert scheme parameterizes closed subschemes of a fixed ambient scheme. A morphism scheme parameterizes maps with both source and target fixed. The connection is provided by the graph of a map:

$$f : X_T \longrightarrow Y_T \quad \rightsquigarrow \quad \Gamma_f \subset X_T \times_T Y_T.$$

The central argument is that being such a graph is an open condition in the appropriate Hilbert scheme. The projective Hilbert scheme existence theorem will be stated, with an explanation of its construction; the passage from Hilbert schemes to morphism schemes will be proved in detail.

1.1 Representable Functors and Universal Families

1.1A The functor of points

Let Sch/S be the category of schemes over S, with S-morphisms as arrows. If H is an S-scheme, its *functor of points* is

$$h_H : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathrm{Sets}, \qquad T \longmapsto \mathrm{Hom}_S(T, H).$$

For a morphism $g : T' \to T$, the induced map is precomposition:

$$h_H(g) : \mathrm{Hom}_S(T, H) \longrightarrow \mathrm{Hom}_S(T', H), \qquad a \longmapsto a \circ g.$$

The superscript op records that arrows are reversed. Moduli problems defined by pulling back families have the same direction.

Definition 1.1.1 (Representable functor)**.** A contravariant functor $F : (\mathrm{Sch}/S)^{\mathrm{op}} \to \mathrm{Sets}$ is *represented* by an S-scheme H if there is a natural isomorphism

$$h_H \xrightarrow{\sim} F.$$

Equivalently, there is an element $u \in F(H)$ such that, for every S-scheme T, the map

$$\mathrm{Hom}_S(T, H) \longrightarrow F(T), \qquad a \longmapsto F(a)(u) \tag{1.1.1}$$

is a bijection. The element u is called the *universal element*. If elements of $F(T)$ are families over T, it is called the *universal family*.

The notation a^*u is often used for $F(a)(u)$. The word "natural" means that these bijections commute with pullback: if $g : T' \to T$, then

$$g^*(a^*u) = (a \circ g)^*u.$$

Thus representability includes compatibility between families, not merely a bijection between closed points and objects.

Proposition 1.1.2 (Universal elements and uniqueness)**.** *A natural transformation $h_H \to F$ is uniquely determined by the image of $\mathrm{id}_H \in h_H(H)$. In particular, a representing scheme with its universal element is unique up to a unique isomorphism preserving that element.*

Proof. If $\eta : h_H \to F$ is natural and $u = \eta_H(\mathrm{id}_H)$, then naturality for $a : T \to H$ gives

$$\eta_T(a) = F(a)\big(\eta_H(\mathrm{id}_H)\big) = F(a)(u).$$

Conversely, this formula defines a natural transformation for any $u \in F(H)$. If (H, u) and (H', u') both represent F, apply their universal properties to u and u'. The resulting maps $H \to H'$ and $H' \to H$ are inverse, because their composites pull back the universal elements to themselves and uniqueness then identifies the composites with the identity maps. □

This is the form of the Yoneda lemma used throughout the lecture.

1.1B Why families contain more information than points

Example 1.1.3 (The affine line)**.** The functor $T \mapsto \Gamma(T, \mathcal{O}_T)$ is represented by $\mathbb{A}^1_S$ when T ranges over S-schemes. A function b corresponds to the map to $\mathbb{A}^1_S$ which pulls back the coordinate x to b.

Example 1.1.4 (An infinitesimal test scheme)**.** Let $D = \operatorname{Spec} k[\epsilon]/(\epsilon^2)$. A D-valued point of $\mathbb{A}^1_k$ is determined by

$$x \longmapsto a + \epsilon b, \qquad a, b \in k.$$

Its closed point is a, while b records a tangent direction. More generally, D-valued points of a k-scheme H with closed point $h \in H(k)$ form its Zariski tangent space $T_{H,h}$. This is why the scheme structure of a parameter space matters for deformation theory.

When a Hilbert scheme represents its functor, a map $T \to \mathrm{Hilb}(X)$ is exactly a flat embedded family over T. An isomorphism between two abstract schemes in X does not identify their Hilbert points unless the embedded subschemes themselves agree.

References. For functors of points and Yoneda, see [5], introductory discussion, pp. 1–2. For the morphism functor and its universal evaluation, compare [1], §2.1C, Theorem 2.1.11, printed pp. 78–79.

1.2 Flatness and Hilbert Polynomials

1.2A Flatness in a family

Definition 1.2.1 (Flat module, sheaf, and morphism)**.** An A-module M is *flat* if $- \otimes_A M$ preserves exact sequences. For $p : V \to T$, an $\mathcal{O}_V$-module $\mathcal{F}$ is *flat over T* if every stalk $\mathcal{F}_v$ is flat over $\mathcal{O}_{T,p(v)}$. The morphism p is flat if $\mathcal{O}_V$ is flat over T.

Flatness over T concerns the base ring, rather than the local rings of V alone. A coherent sheaf can be flat over T without being locally free on V. For example, if $T = \operatorname{Spec} k$, every k-vector space is flat, so every coherent sheaf on a k-scheme is flat over k.

Two properties will be used repeatedly. Flatness is preserved by arbitrary base change. Also, if

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow 0 \tag{1.2.1}$$

is exact and $\mathcal{G}$ is flat over T, restriction to a fiber remains exact. Locally, the only possible failure of left exactness is measured by $\operatorname{Tor}_1^A(\mathcal{G}, \kappa(t))$, which vanishes by flatness.

Example 1.2.2 (Two points colliding)**.** Set $T = \operatorname{Spec} k[t]$ and

$$V = \operatorname{Spec}\big(k[t,x]/(x(x-t))\big).$$

Its coordinate ring is free over $k[t]$ with basis $1, x$: division by the monic polynomial $x^2 - tx$ gives a unique expression $a(t) + b(t)x$. Thus $V \to T$ is finite flat of degree two. At $t = a \neq 0$, the fiber consists of the two reduced points $x = 0$ and $x = a$. At $t = 0$, the fiber is the double point

$$\operatorname{Spec} k[x]/(x^2).$$

The number of points in the underlying topological space drops, but the scheme-theoretic length stays equal to two. The same family is a closed subscheme of $\mathbb{P}^1_T$, given in homogeneous coordinates by $X(X - tY) = 0$; it has no points on $Y = 0$.

Example 1.2.3 (A point appearing only in one fiber)**.** Inside $\mathbb{P}^1_T$, with $T = \operatorname{Spec} k[t]$, take the subscheme given by $X = 0$ and $t = 0$. Its fiber is empty for $t \neq 0$ and a reduced point for $t = 0$. Its structure module is $k[t]/(t)$, which is not flat over $k[t]$: tensoring the injective multiplication map $k[t] \xrightarrow{t} k[t]$ with this module gives the zero map on a nonzero module.

References. See [3], III, §9 for flatness, and [6], Tag 051D for base change and flatness. Fiberwise exactness follows from the vanishing of the corresponding Tor_1, as explained above.

1.2B The polynomial of a coherent sheaf

Let V be projective over a field K, let L be an ample line bundle on V, and let $\mathcal{F}$ be coherent. Write $\mathcal{F}(m) = \mathcal{F} \otimes L^{\otimes m}$, using the dual of L when $m < 0$.

Definition 1.2.4 (Hilbert polynomial)**.** The *Hilbert polynomial* of $\mathcal{F}$ with respect to L is the polynomial $P_{\mathcal{F},L} \in \mathbb{Q}[m]$ such that

$$P_{\mathcal{F},L}(m) = \chi(V, \mathcal{F}(m)) = \sum_{i \geq 0} (-1)^i \dim_K H^i(V, \mathcal{F}(m)). \tag{1.2.2}$$

For projective V, the right-hand side is polynomial in m for all integers m. In particular, for $m \gg 0$, Serre vanishing gives

$$P_{\mathcal{F},L}(m) = h^0(V, \mathcal{F}(m)).$$

The Hilbert polynomial of a closed subscheme $W \subset V$ is that of $\mathcal{O}_W$, with polarization $L|_W$.

The eventual dimension of sections determines the polynomial uniquely. At small values of m, however, h^0 need not equal the polynomial; the Euler characteristic is the correct invariant. Cohomology groups in (1.2.2) are finite dimensional, by properness and coherence.

Proposition 1.2.5 (Numerical polynomials)**.** *If $P \in \mathbb{Q}[m]$ takes integer values on all sufficiently large integers, then it has a unique expansion*

$$P(m) = \sum_{i=0}^{r} a_i \binom{m}{i}, \qquad a_i \in \mathbb{Z}, \tag{1.2.3}$$

where $\binom{m}{0} = 1$ and $\binom{m}{i} = m(m-1)\cdots(m-i+1)/i!$ for $i > 0$. In particular, P is integer valued on every integer.

Proof. Let $\Delta P(m) = P(m+1) - P(m)$. The binomial polynomials satisfy

$$\Delta \binom{m}{i} = \binom{m}{i-1} \quad (i \geq 1),$$

and Newton's formula gives $P(m) = \sum_i \Delta^i P(0) \binom{m}{i}$. To check that the coefficients are integral without initially assuming integrality at 0, choose an integer M beyond the given bound and expand instead at M:

$$P(m) = \sum_i \Delta^i P(M) \binom{m-M}{i}.$$

Every coefficient in this expression is integral and every binomial term is integer valued on $\mathbb{Z}$. Thus P is integer valued on all of $\mathbb{Z}$, and hence $a_i = \Delta^i P(0) \in \mathbb{Z}$. Uniqueness follows because the binomial polynomials have distinct degrees and nonzero leading coefficients. □

An integer-valued polynomial need not have integer coefficients in the monomial basis. For example, $(m^2 + 3m + 2)/2 = \binom{m+2}{2}$ is integer valued.

References. For existence of Hilbert polynomials and their degree, see [3], I, §7, and III, §5 for cohomology of projective space. For a direct sheaf-theoretic formulation, see [6], Tag 08AC and Tag 08AD.

1.2C Examples and the numerical information retained

Example 1.2.6 (Projective space and linear subspaces)**.** With the standard polarization,

$$P_{\mathbb{P}^r}(m) = \binom{m+r}{r}.$$

Indeed, for $m \geq 0$, the sections are degree-m homogeneous polynomials in $r+1$ variables, and their number is the displayed binomial coefficient. A linear $\mathbb{P}^r \subset \mathbb{P}^N$ has the same Hilbert polynomial.

Example 1.2.7 (Zero-dimensional schemes)**.** If W is a finite K-scheme of length $\ell = \dim_K H^0(W, \mathcal{O}_W)$, then $P_W(m) = \ell$. An invertible sheaf on each Artinian local component is free of rank one, so twisting does not change this dimension. Higher cohomology vanishes because W is affine. Distinct reduced points and a nonreduced scheme with the same length therefore have the same polynomial.

Example 1.2.8 (Smooth curves)**.** If C is a smooth connected projective curve of genus g over an algebraically closed field, and L has degree $d > 0$, Riemann–Roch gives

$$P_{C,L}(m) = dm + 1 - g.$$

For an embedded smooth rational curve of degree d, the polynomial is $dm+1$. Thus a line, a conic, and a twisted cubic have polynomials $m+1$, $2m+1$, and $3m+1$, respectively. On a general projective curve the corresponding constant term is $1 - p_a(C)$, where $p_a(C) = 1 - \chi(C, \mathcal{O}_C)$ is the arithmetic genus.

Example 1.2.9 (Hypersurfaces)**.** A degree-d hypersurface $D \subset \mathbb{P}^N$, with $N \geq 1$, has the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^N}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^N} \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

Taking Euler characteristics after twisting gives

$$P_D(m) = \binom{m+N}{N} - \binom{m+N-d}{N}.$$

For a plane conic this is $2m+1$, including when the equation defines two lines or a double line. For a plane cubic it is $3m$.

Remark 1.2.10 (The Hilbert polynomial does not determine the subscheme)**.** A twisted cubic in $\mathbb{P}^3$ has polynomial $3m + 1$. A plane cubic together with a reduced point outside its plane has the same polynomial, since the point adds 1. These are very different subschemes. Fixing a Hilbert polynomial gives a bounded moduli problem, but does not by itself impose smoothness, connectedness, or rationality.

For a nonempty projective scheme W with an ample polarization,

$$\deg P_W = \dim W.$$

If W has dimension r, the leading coefficient is $\deg_L(W)/r!$, where the degree includes the scheme-theoretic multiplicities of its top-dimensional components. The polynomial therefore determines the dimension and total polarized degree.

References. See [3], I, §7 for dimension and degree, and IV, §1 for the curve calculation by Riemann–Roch. For arithmetic genus, compare III, §9, especially the discussion of invariants in flat projective families.

1.2D Constancy in flat families

Let $p : V \to T$ be projective, let T be noetherian, let L be relatively ample, and let $\mathcal{F}$ be coherent and flat over T. Put

$$V_t = V \times_T \operatorname{Spec} \kappa(t), \qquad \mathcal{F}_t = \mathcal{F} \otimes_{\mathcal{O}_T} \kappa(t).$$

The ambient morphism p need not be flat: it is the sheaf $\mathcal{F}$ whose flatness is assumed.

Theorem 1.2.11 (High twists and base change)**.** *There is an integer m_0 such that, for $m \geq m_0$:*

(i) $R^i p_* \mathcal{F}(m) = 0$ for $i > 0$, and $H^i(V_t, \mathcal{F}_t(m)) = 0$ for all $t \in T$ and $i > 0$;

(ii) $p_* \mathcal{F}(m)$ is a finite locally free $\mathcal{O}_T$-module;

(iii) its formation commutes with arbitrary base change. In particular,

$$p_*\mathcal{F}(m) \otimes_{\mathcal{O}_T} \kappa(t) \xrightarrow{\sim} H^0(V_t, \mathcal{F}_t(m)). \tag{1.2.4}$$

This combines relative Serre vanishing with cohomology and base change. The noetherian, hence quasi-compact, base allows one uniform m_0. On a merely locally noetherian base the assertion holds locally on the base.

References. For relative Serre vanishing, see [3], III, §8; for cohomology and base change, see III, §12, Theorem 12.11. A formulation which explicitly gives arbitrary base change and finite local freeness after higher direct images vanish is [6], Tag 0D4E. These are external cohomological results used here.

Corollary 1.2.12 (Local constancy of the Hilbert polynomial)**.** *The function $t \mapsto P_{\mathcal{F}_t, L_t}$ is locally constant on T. If T is connected, there is a single polynomial $P_{\mathcal{F},L}$ and*

$$\operatorname{rank} p_*\mathcal{F}(m) = P_{\mathcal{F},L}(m) \qquad (m \gg 0).$$

Proof. For sufficiently large m, Theorem 1.2.11 identifies $P_{\mathcal{F}_t,L_t}(m)$ with the rank of the vector bundle $p_*\mathcal{F}(m)$ at t, so this value is locally constant. There is a finite bound r for the degrees of the fiber polynomials, for example the dimension of a projective space containing V locally over the base. Near a fixed point, the ranks for $r+1$ consecutive large values of m are simultaneously constant. These $r+1$ values determine a polynomial of degree at most r, proving local constancy of the polynomial itself. On a connected base a locally constant function is constant. □

Proposition 1.2.13 (Invariance under arbitrary base change)**.** *Let $g : T' \to T$ be any morphism, put $V' = V \times_T T'$, and let $\mathcal{F}'$ and L' be the pullbacks. For $t' \in T'$ with image $t \in T$,*

$$P_{\mathcal{F}'_{t'}, L'_{t'}} = P_{\mathcal{F}_t, L_t}.$$

Consequently, if the family over T has a single Hilbert polynomial P, the pullback has that same polynomial on every fiber, whether or not T' is connected.

Proof. The residue fields have an extension $\kappa(t) \hookrightarrow \kappa(t')$, and

$$V'_{t'} \simeq V_t \times_{\kappa(t)} \kappa(t'), \qquad \mathcal{F}'_{t'} \simeq \mathcal{F}_t \otimes_{\kappa(t)} \kappa(t').$$

Cohomology commutes with a field extension, which is a flat base change. For every integer m and every i this gives

$$H^i(V'_{t'}, \mathcal{F}'_{t'}(m)) \simeq H^i(V_t, \mathcal{F}_t(m)) \otimes_{\kappa(t)} \kappa(t').$$

Extension of scalars preserves the dimension of a finite-dimensional vector space. The Euler characteristics, and hence the polynomials, agree. Flatness of $\mathcal{F}'$ over T' follows independently from stability of flatness under base change. □

Remark 1.2.14 (Two different base change statements)**.** Flat base change for cohomology requires the *base change map* to be flat; it does not mean that the original projective morphism must be flat. In Proposition 1.2.13, the relevant flat map is $\operatorname{Spec}\kappa(t') \to \operatorname{Spec}\kappa(t)$. The map $T' \to T$ itself may be arbitrary. Also, a connected T does not imply a connected T'.

References. For flat base change, see [3], III, Proposition 9.3. The fiberwise proof above explains why no connectedness assumption on T' is required.

Example 1.2.15 (Constant polynomials alone need not give flatness)**.** Let $T = \operatorname{Spec} A$ with $A = k[\epsilon]/(\epsilon^2)$, take $V = T$ with its identity morphism, and let $\mathcal{F}$ be the sheaf associated to $A/(\epsilon)$. There is only one point of T, so its fiber Hilbert polynomial is constantly 1. Nevertheless $\mathcal{F}$ is not flat over A.

Indeed, tensor the inclusion $(\epsilon) \hookrightarrow A$ with $A/(\epsilon)$. The source becomes $(\epsilon)/(\epsilon^2) \simeq k$, whereas the induced map to $A/(\epsilon)$ is zero. Tensoring has destroyed injectivity. Numerical fiber data cannot detect all infinitesimal structure of a nonreduced base.

1.2E Exact sequences and finite differences

Euler characteristic is additive in a short exact sequence of coherent sheaves. For a family, flatness of the quotient ensures that this exact sequence remains exact on fibers.

Proposition 1.2.16 (A hyperplane-type exact sequence)**.** *In the preceding flat family, suppose there is an exact sequence*

$$0 \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow 0$$

and that $\mathcal{G}$ is flat over T. On a connected base,

$$P_{\mathcal{G}}(m) = P_{\mathcal{F}}(m) - P_{\mathcal{F}}(m-1). \tag{1.2.5}$$

If $P_{\mathcal{F}}(m) = \sum_{i=0}^{r} a_i \binom{m}{i}$, then

$$P_{\mathcal{G}}(m) = \sum_{i=1}^{r} a_i \binom{m-1}{i-1} = \sum_{j=0}^{r-1} a_{j+1} \binom{m-1}{j}.$$

Proof. Restrict the exact sequence to a fiber; it remains exact because the quotient is flat over the base. Twist by $L_t^{\otimes m}$ and take Euler characteristics. This gives (1.2.5). The coefficient formula is Pascal's identity

$$\binom{m}{i} - \binom{m-1}{i} = \binom{m-1}{i-1}, \qquad i \geq 1.$$

The constant term a_0 disappears, and reindexing uses $j = i - 1$. □

Example 1.2.17 (A hyperplane section of a curve)**.** For a smooth curve C of degree d, a hyperplane whose defining section is not identically zero on C gives

$$0 \longrightarrow \mathcal{O}_C(-1) \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{O}_{C \cap H} \longrightarrow 0.$$

Since $P_C(m) = dm + 1 - g$, its finite difference is d. Thus the intersection has length d, even when it is nonreduced. In a relative hyperplane-section argument, the injectivity of the first arrow and flatness of the quotient must both be checked.

1.3 The Hilbert Functor

1.3A Embedded flat families

Let $X \to S$ be a separated morphism of finite type. In the projective case used below, every closed subscheme of X_T is automatically proper over T.

Definition 1.3.1 (Hilbert functor)**.** The *Hilbert functor* of X/S is the contravariant functor

$$\mathcal{H}ilb_{X/S} : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathrm{Sets}$$

whose value on T is the set of closed subschemes $V \subset X_T$ such that $V \to T$ is proper, flat, and of finite presentation. For $g : T' \to T$, its pullback map sends V to

$$V_{T'} = V \times_T T' \subset X_{T'}.$$

The finite-presentation condition is automatic for closed subschemes when the ambient scheme and the test scheme are noetherian. It is included to make the functor correct for arbitrary test schemes. Properness, flatness, and finite presentation are preserved by base change, so this is indeed a functor.

Remark 1.3.2 (Subschemes as quotients)**.** A closed subscheme $V \subset X_T$ corresponds to a quasi-coherent ideal $\mathcal{I}_V \subset \mathcal{O}_{X_T}$ and the quotient

$$\mathcal{O}_{X_T} \twoheadrightarrow \mathcal{O}_V = \mathcal{O}_{X_T}/\mathcal{I}_V.$$

Conversely, the kernel of any surjective $\mathcal{O}_{X_T}$-module map from $\mathcal{O}_{X_T}$ is an ideal and determines a closed subscheme. The quotient must be retained together with its quotient map: forgetting that map forgets the embedding. This realizes the Hilbert functor as the special case of the Quot functor in which the original sheaf is $\mathcal{O}_X$.

1.3B Fixing a Hilbert polynomial

Now assume $X \to S$ is projective and choose a relatively ample line bundle L on X.

Definition 1.3.3 (Fixed-polynomial Hilbert functor)**.** For $P \in \mathbb{Q}[m]$, the subfunctor $\mathcal{H}ilb^{P}_{X/S}$ consists of families $V \subset X_T$ for which every fiber V_t has Hilbert polynomial P with respect to $L_t|_{V_t}$. When the polarization needs to be emphasized, write $\mathcal{H}ilb^{P,L}_{X/S}$.

Proposition 1.2.13 makes this condition stable under arbitrary base change. If P is not a polynomial of any subscheme of a fiber, the functor may simply be empty.

On a connected test scheme, every flat family has one Hilbert polynomial, and one may write the set-theoretic disjoint union

$$\mathcal{H}ilb_{X/S}(T) = \coprod_{P} \mathcal{H}ilb^{P}_{X/S}(T) \qquad (T \text{ connected and nonempty}). \tag{1.3.1}$$

The case of a disconnected base requires more care. A family determines a locally constant polynomial function and an open and closed decomposition

$$T = \coprod_{P} T_P,$$

with a family of polynomial P over each T_P. Different parts of the same test scheme can therefore map to different polynomial components of the representing scheme.

Remark 1.3.4 (A coproduct of schemes is not a pointwise union on every test scheme)**.** If $H = \coprod_P H_P$, a map $T \to H$ is a choice of open and closed pieces T_P together with maps $T_P \to H_P$. It need not send all of T into one H_P. Consequently (1.3.1) must not be asserted for arbitrary disconnected T as an ordinary disjoint union of sets. The empty test scheme also has one map to any scheme, including a coproduct.

1.3C Length one and base change

Lemma 1.3.5 (Finite flat families of rank one)**.** *A finite, flat morphism of finite presentation $p : V \to T$ whose fibers have length one is an isomorphism.*

Proof. Such a morphism is finite locally free, and its algebra $p_*\mathcal{O}_V$ is locally free of rank one. The unit map

$$\mathcal{O}_T \longrightarrow p_*\mathcal{O}_V$$

is an isomorphism on every residue-field fiber, since a nonzero one-dimensional unital algebra over a field is that field. In local bases it is multiplication by an element which is a unit in every relevant local ring. Thus it is an isomorphism of algebras, and taking relative spectra gives $V \simeq T$. □

Proposition 1.3.6 (The Hilbert scheme of one point)**.** *If $X \to S$ is projective, then*

$$\mathcal{H}ilb^1_{X/S} \simeq h_X.$$

In particular, the representing scheme is $\mathrm{Hilb}^1(X/S) \simeq X$.

Proof. For $a : T \to X$, separatedness of X/S makes its graph

$$\Gamma_a \subset X \times_S T$$

a closed subscheme isomorphic to T. Its projection to T is the identity under this identification, hence proper and flat; every fiber has polynomial 1. This gives $h_X(T) \to \mathcal{H}ilb^1_{X/S}(T)$.

Conversely, let $V \subset X_T$ be such a family. Since each fiber has a nonzero constant Hilbert polynomial, the dimension-degree relation shows that it is zero dimensional. Its length is the value of the polynomial, namely 1. Therefore $V \to T$ is proper and quasi-finite, hence finite. Flatness and finite presentation make it finite locally free of rank one. By Lemma 1.3.5, it is an isomorphism. Its inverse followed by $V \hookrightarrow X_T \to X$ is the desired morphism $T \to X$. These two operations are inverse and commute with pullback. □

Under this identification, the universal family is the diagonal $\Delta_X \subset X \times_S X$.

Remark 1.3.7 (Why the flatness hypothesis matters)**.** A finite morphism which has generic degree one need not be an isomorphism. For example, the normalization of a cuspidal curve is finite and birational but is not an isomorphism at the cusp. Lemma 1.3.5 uses *finite local freeness of rank one*, which follows here from the flatness already present in the Hilbert functor.

Proposition 1.3.8 (Base change of the Hilbert functor)**.** *Let $S' \to S$ be a morphism. Restricting the functor to S'-schemes gives*

$$\mathcal{H}ilb^P_{X_{S'}/S'} \simeq \mathcal{H}ilb^P_{X/S} \times_{h_S} h_{S'}.$$

If $\mathcal{H}ilb^P_{X/S}$ is represented by H_P, then the left-hand functor is represented by $H_P \times_S S'$.

Proof. For an S'-scheme T, the canonical identification

$$(X \times_S S') \times_{S'} T \simeq X \times_S T$$

identifies the ambient schemes and hence the embedded families. The flatness, properness, and fiber-polynomial conditions agree. The assertion about representing schemes then follows from the universal property and Yoneda. □

References. For finite local freeness, see [6], Tag 02K9. For Hilbert schemes of points, see Tag 0B94. The general Hilbert functor, including finite-presentation conventions, is discussed in Tag 0CZX; the projective scheme construction is the classical theorem below.

1.4 Existence of Hilbert Schemes

1.4A The existence theorem and its universal property

Theorem 1.4.1 (Grothendieck's Hilbert scheme theorem)**.** *Let S be noetherian, let $X \to S$ be projective, and let L be a relatively very ample line bundle. For each $P \in \mathbb{Q}[m]$, the functor $\mathcal{H}ilb^{P,L}_{X/S}$ is represented by a projective S-scheme*

$$H_P = \mathrm{Hilb}^{P,L}(X/S).$$

There is a universal closed subscheme

$$\mathcal{U}_P \subset X \times_S H_P$$

which is flat over H_P and has fiber Hilbert polynomial P. For every S-scheme T and every $V \in \mathcal{H}ilb^{P,L}_{X/S}(T)$, there is a unique classifying map $a : T \to H_P$ for which

$$V = (\mathrm{id}_X \times a)^{-1}\mathcal{U}_P = \mathcal{U}_P \times_{H_P} T \tag{1.4.1}$$

as closed subschemes of X_T.

Remark 1.4.2 (Ample versus very ample)**.** The existence theorem is also valid for a relatively ample polarization. Its standard construction uses a sufficiently large relatively very ample tensor power, locally on the base. Replacing L by $L^{\otimes e}$ replaces the polynomial by $P(em)$, since $\mathcal{F} \otimes (L^{\otimes e})^{\otimes m} = \mathcal{F}(em)$. No information is lost: the substitution $P(m) \mapsto P(em)$ is injective on $\mathbb{Q}[m]$. In particular, the phrase "relatively ample" in the definition of the polynomial does not supply a projective embedding by L itself.

The universal property can be displayed by the cartesian diagram

$$\begin{array}{ccc} V & \longrightarrow & \mathcal{U}_P \\ \downarrow & & \downarrow \\ T & \xrightarrow{\ a\ } & H_P. \end{array}$$

It is (1.4.1), rather than merely a list of closed points of H_P, which is used to construct other parameter spaces.

Allowing all polynomials gives a locally noetherian scheme

$$\mathrm{Hilb}(X/S) = \coprod_P \mathrm{Hilb}^{P,L}(X/S)$$

representing the full Hilbert functor, with the interpretation of disconnected test schemes explained in §1.3. Each fixed-polynomial piece is projective over S. The full Hilbert scheme generally has infinitely many pieces and need not be of finite type or projective over S.

References. For the original theorem, see [2], Exposé 221. A detailed accessible construction is [5], §5, Theorem 5.1, pp. 24–28, applied to $E = \mathcal{O}_X$. See also [4], Chapter I, §1.

1.4B Why a Grassmannian enters the construction

Here is the mechanism over a field. Choose an embedding $X \subset \mathbb{P}^N_k$ and use its hyperplane bundle. For an embedded subscheme $V \subset X$, a sufficiently large twist gives an exact sequence

$$0 \longrightarrow H^0(\mathbb{P}^N, \mathcal{I}_V(m)) \longrightarrow H^0(\mathbb{P}^N, \mathcal{O}(m)) \longrightarrow H^0(V, \mathcal{O}_V(m)) \longrightarrow 0. \tag{1.4.2}$$

The last space has dimension $P(m)$. Thus V gives a quotient of dimension $P(m)$ of one fixed vector space, and hence a point of a Grassmannian.

The essential assertion is a *uniform* bound on m for all V with the prescribed polynomial. Serre vanishing for each individual V does not give such a bound. Castelnuovo–Mumford regularity and boundedness of quotients supply it.

Definition 1.4.3 (Regularity)**.** A coherent sheaf $\mathcal{F}$ on $\mathbb{P}^N_k$ is *m-regular* if

$$H^i(\mathbb{P}^N, \mathcal{F}(m-i)) = 0 \qquad (i > 0).$$

Regularity controls both generation by global sections and cohomology of higher twists.

For a fixed Hilbert polynomial, one can choose m for which all relevant ideal sheaves are m-regular. Their degree-m sections then generate the ideals as sheaves, so the quotient in (1.4.2) determines the embedded subscheme. In a flat family, Theorem 1.2.11 replaces the quotient vector space by a vector bundle of rank $P(m)$, and the relative Grassmannian parameterizes these quotients.

Not every quotient in the Grassmannian comes from a family with the required polynomial. Compatibility with multiplication in the homogeneous coordinate ring, containment in X, and flatness must still be imposed. A standard construction first obtains a locally closed locus using flattening stratification, and then proves properness by the valuative criterion; the resulting immersion into the Grassmannian is closed. This is the substantive construction behind Theorem 1.4.1, and it is not replaced by simply choosing an arbitrary degree-m subspace.

For orientation, the properness step over a discrete valuation ring R has a concrete interpretation. If $V_K \subset X_K$ is a generic-fiber subscheme, its scheme-theoretic closure in X_R has a structure sheaf which embeds locally into the generic-fiber structure sheaf. It is therefore R-torsion free, hence R-flat. This is its flat limit. Uniqueness follows because a flat quotient has no R-torsion: its ideal is saturated with respect to a uniformizer. This

is why projective Hilbert schemes can retain nonreduced and reducible limits of smooth subschemes.

References. For regularity, see [5], §2; for flattening stratification, §4, Theorem 4.3; for the construction and valuative argument, §5. A uniform regularity statement is also [6], Tag 08AG. The full proof of Hilbert scheme existence is external to this lecture.

1.5 Morphism Schemes as Open Hilbert Subschemes

1.5A The morphism functor and graphs

Definition 1.5.1 (Morphism functor)**.** For S-schemes X and Y, define

$$\mathcal{M}or_{X,Y/S}(T) = \mathrm{Hom}_T(X_T, Y_T).$$

Base change of a map along $T' \to T$ makes this a contravariant functor on Sch/S. Its representing scheme, when it exists, is denoted by $\mathrm{Mor}_S(X,Y)$ or $\mathrm{Hom}_S(X,Y)$.

The set $\mathrm{Hom}_T(X_T, Y_T)$ should be distinguished from the scheme $\mathrm{Mor}_S(X,Y)$. The latter represents all such sets simultaneously. There is also a canonical identification $\mathrm{Hom}_T(X_T, Y_T) = \mathrm{Hom}_S(X_T, Y)$.

Lemma 1.5.2 (Closed graphs)**.** *If $Y \to S$ is separated, the graph of a T-morphism $f : X_T \to Y_T$ is a closed subscheme*

$$\Gamma_f \subset X_T \times_T Y_T \simeq (X \times_S Y)_T.$$

Its first projection $\Gamma_f \to X_T$ is an isomorphism.

Proof. The graph map $x \mapsto (x, f(x))$ is a base change of the diagonal of Y_T/T, which is a closed immersion by separatedness. The first projection is inverse to the graph map onto its image. □

If X is projective and flat over S, then X_T is proper, flat, and of finite presentation over T. Its graph has these same properties. Therefore graphs give a natural transformation

$$\mathcal{M}or_{X,Y/S} \longrightarrow \mathcal{H}ilb_{X\times_S Y/S}, \qquad f \longmapsto \Gamma_f. \tag{1.5.1}$$

It is injective on every test scheme: the first projection and the embedded graph recover f.

1.5B An isomorphism condition in a proper family

The following result is the key to identifying the image of (1.5.1) as an open subfunctor.

Lemma 1.5.3 (An isomorphism persists near a fiber)**.** *Let B be noetherian, let $A \to B$ and $C \to B$ be proper morphisms, and suppose A is flat over B. Let $q : A \to C$ be a B-morphism. If $q_b : A_b \to C_b$ is an isomorphism for some $b \in B$, then there is an open neighborhood $B' \subset B$ of b for which $q_{B'}$ is an isomorphism.*

Proof. First, q is proper. Indeed, its graph in $A \times_B C$ is closed because C/B is separated, and $A \times_B C \to C$ is proper.

Since q_b is an isomorphism, q is quasi-finite at every point of A_b. The quasi-finite locus is open. Its closed complement has closed image in B, because $A \to B$ is proper, and that image does not contain b. Remove this image. Now q is proper and quasi-finite, hence finite.

Consider the coherent algebra map

$$\mathcal{O}_C \longrightarrow q_*\mathcal{O}_A.$$

For a finite morphism, pushforward commutes with base change, so its restriction to C_b is the corresponding isomorphism for q_b. The cokernel therefore vanishes at all points of C_b, by Nakayama's lemma. Its support has closed image in B by properness of C/B. Shrinking B again makes the map surjective everywhere. Thus q is a closed immersion, with a coherent kernel $\mathcal{J}$ in

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_C \longrightarrow q_*\mathcal{O}_A \longrightarrow 0.$$

The last sheaf is flat over B, since A/B is flat and q is finite. Hence restriction to C_b remains left exact. Since the restricted algebra map is an isomorphism, $\mathcal{J} \otimes \kappa(b) = 0$. Nakayama's lemma gives vanishing of $\mathcal{J}$ along C_b; properness of C/B allows another shrinking to remove its support. The algebra map is now an isomorphism, and so is q. □

Corollary 1.5.4 (The open isomorphism locus)**.** *Under the assumptions of Lemma 1.5.3, the subset*

$$W = \{b \in B \mid q_b : A_b \to C_b \text{ is an isomorphism}\}$$

is open, and $q_W : A_W \to C_W$ is an isomorphism.

Proof. Every point of W has an open neighborhood on which q is an isomorphism by the lemma. These neighborhoods lie in W and cover it. The local inverse morphisms agree on overlaps because each is inverse to q, so they glue. □

Remark 1.5.5 (Testing after extending the residue field)**.** If $K \subset K'$ is a field extension and a K-morphism becomes an isomorphism after base change to K', then it was already an isomorphism. This follows from faithfully flat descent of isomorphisms, since $\operatorname{Spec} K' \to \operatorname{Spec} K$ is faithfully flat. This observation is needed when a classifying map sends a point t to a point h: their residue fields need not be equal.

References. Lemma 1.5.3 is [6], Tag 0G0Q. For the openness and properness ingredients, see [3], II, §4 and III, §10, or [6], the proof linked from that tag. Descent of isomorphisms is treated in Tag 023R.

1.5C Construction of the representing scheme

Theorem 1.5.6 (Morphism schemes)**.** *Let X and Y be projective over the noetherian scheme S, and assume that X is flat over S. Then $\mathcal{M}or_{X,Y/S}$ is represented by an open subscheme*

$$\operatorname{Mor}_S(X, Y) \subset \operatorname{Hilb}(X \times_S Y/S).$$

It is locally of finite type over S. There is a universal evaluation morphism

$$\operatorname{ev} : X \times_S \operatorname{Mor}_S(X, Y) \longrightarrow Y$$

whose pullback along $a : T \to \operatorname{Mor}_S(X, Y)$ is the corresponding family of maps $X_T \to Y$.

Proof. Write $H = \mathrm{Hilb}(X \times_S Y/S)$ and let

$$\mathcal{U} \subset (X \times_S Y) \times_S H$$

be its universal family. Restricting the first projection gives a morphism

$$q : \mathcal{U} \longrightarrow X_H = X \times_S H.$$

Both $\mathcal{U}$ and X_H are proper over H, and $\mathcal{U}/H$ is flat. Each fixed-polynomial component of H is noetherian. Applying Corollary 1.5.4 on these components gives an open subscheme

$$W = \{h \in H \mid q_h : \mathcal{U}_h \xrightarrow{\sim} X_{\kappa(h)}\}. \tag{1.5.2}$$

The restricted map $q_W : \mathcal{U}_W \to X_W$ is an isomorphism.

Step 1: a point of W gives a family of morphisms. Let $a : T \to W$. Pull back the universal family. Base change of q_W identifies $\mathcal{U}_T$ with X_T, and the second projection gives

$$X_T \xrightarrow{\sim} \mathcal{U}_T \hookrightarrow X_T \times_T Y_T \xrightarrow{\mathrm{pr}_2} Y_T.$$

This is a T-morphism $f_a : X_T \to Y_T$. Its graph is exactly the embedded family $\mathcal{U}_T$, because the first projection is the chosen identification with X_T.

Step 2: a family of morphisms gives a point of W. Let $f : X_T \to Y_T$. Its graph is a Hilbert family, so the universal property gives a unique classifying map $a_f : T \to H$ such that $\Gamma_f = \mathcal{U} \times_H T$ as embedded subschemes. For every $t \in T$, set $h = a_f(t)$. Then

$$\mathcal{U}_h \times_{\kappa(h)} \kappa(t) \simeq (\Gamma_f)_t, \qquad X_{\kappa(h)} \times_{\kappa(h)} \kappa(t) \simeq X_t.$$

Under these identifications, the base change of q_h is the first projection $(\Gamma_f)_t \to X_t$, an isomorphism. Remark 1.5.5 therefore implies that q_h is an isomorphism. Hence $a_f(t) \in W$ for every t.

Since $W \subset H$ is open, a morphism whose entire image lies in W factors uniquely through W. Thus $a_f : T \to W$ is defined even when T is nonreduced or not noetherian. Both constructions commute with base change, and recovering a map from its graph makes them inverse. We obtain the natural bijection

$$\mathrm{Hom}_S(T, W) \xrightarrow{\sim} \mathrm{Hom}_T(X_T, Y_T).$$

This proves representability. The universal evaluation is the composite

$$X_W \xrightarrow{q_W^{-1}} \mathcal{U}_W \hookrightarrow X_W \times_W Y_W \longrightarrow Y_W \longrightarrow Y.$$

Finally, H is a disjoint union of schemes projective over S, hence locally of finite type over S, and the same holds for its open subscheme W. □

Remark 1.5.7 (Where the hypotheses enter)**.** Separatedness of Y/S makes graphs closed. Properness of X/S makes the graph proper over the parameter scheme. Flatness of X/S makes it a flat Hilbert family. Hilbert scheme existence produces H, and the proper-family isomorphism lemma produces its open graph locus. When $S = \mathrm{Spec}\, k$, the flatness assumption on X is automatic.

Remark 1.5.8 (A quasi-projective target)**.** The theorem remains true for quasi-projective Y/S. Embed Y as an open subscheme of a projective compactification $\overline{Y}$. Construct $\mathrm{Mor}_S(X, \overline{Y})$ first. In its universal family, the set of parameters whose maps meet $\overline{Y} \setminus Y$ is the image of a closed subset of $X \times_S \mathrm{Mor}_S(X, \overline{Y})$. This image is closed because X/S is proper. Its complement is the open subscheme representing maps to Y. Equivalently, the graph construction uses the Hilbert functor of subschemes with proper support in $X \times_S Y$.

References. The graph construction is [5], §6, Theorem 6.6, pp. 30–31; Theorem 6.5 gives an alternative proof of the open isomorphism condition. Compare [1], Theorem 2.1.11 and the graph-polynomial discussion on printed pp. 78–79.

1.6 Examples and the Passage to Rational Curves

1.6A The polynomial of a graph

Let X and Y be projective over k, with very ample line bundles A and B. Use the product polarization

$$A \boxtimes B = \mathrm{pr}_X^* A \otimes \mathrm{pr}_Y^* B$$

on $X \times_k Y$. For $f : X \to Y$, the first projection identifies its graph with X, and the product polarization restricts to $A \otimes f^* B$. Hence

$$P_{\Gamma_f}(m) = \chi\big(X, A^{\otimes m} \otimes f^* B^{\otimes m}\big). \tag{1.6.1}$$

Fixing a graph polynomial gives an open subscheme of a projective Hilbert scheme, so a scheme of finite type which is quasi-projective over k. Taking all graph polynomials gives the entire morphism scheme, which generally is only locally of finite type.

Example 1.6.1 (A point as source)**.** If $X = \operatorname{Spec} k$, then $\mathrm{Mor}_k(X, Y) \simeq Y$. Its graph construction is precisely the identification $\mathrm{Hilb}^1(Y/k) \simeq Y$ from Proposition 1.3.6.

Example 1.6.2 (Degree-d maps from the projective line)**.** Take $X = \mathbb{P}^1$, $Y = \mathbb{P}^N$, $A = \mathcal{O}_{\mathbb{P}^1}(1)$, and $B = \mathcal{O}_{\mathbb{P}^N}(1)$. If $f^* B = \mathcal{O}_{\mathbb{P}^1}(d)$, then

$$P_{\Gamma_f}(m) = (d+1)m + 1.$$

The $d+1$ appears because the polarization on the graph is the *product* polarization. The polynomial is not $dm + 1$, which would be the polynomial of a smooth degree-d image curve with its target polarization when f is an embedding.

Concretely, a degree-d map is an $(N+1)$-tuple of degree-d binary forms $F_0, \dots, F_N$ with no common zero, up to common nonzero scalar:

$$[u : v] \longmapsto [F_0(u, v) : \cdots : F_N(u, v)].$$

Let $E = H^0(\mathbb{P}^1, \mathcal{O}(d))^{\oplus(N+1)}$. In the projective space of lines $\mathbb{P}(E)$, the no-common-zero condition is open: the bad tuples are the image of the closed incidence locus

$$\{(p, [F]) \in \mathbb{P}^1 \times \mathbb{P}(E) \mid F_i(p) = 0 \text{ for every } i\},$$

under the proper projection to $\mathbb{P}(E)$. This open subscheme represents $\mathrm{Mor}_d(\mathbb{P}^1, \mathbb{P}^N)$. Over a general base, the forms can be written after locally trivializing the line bundle

from that base; projective coefficient space retains precisely the transition scalars needed to glue them.

If it is nonempty, this space is smooth of dimension

$$(N+1)(d+1)-1.$$

For $d=0$ it is $\mathbb{P}^N$, the space of constant maps. For $N=1, d=1$, it is the open subset of projective 2×2 matrices with nonzero determinant, namely PGL_2. This also shows that a morphism scheme need not be proper, even though each fixed-polynomial Hilbert scheme is projective.

Example 1.6.3 (A limit of graphs leaves the graph locus)**.** On $\mathbb{P}^1\times\mathbb{P}^1$, with coordinates $[u:v]$ and $[x:y]$, consider the family of divisors

$$xv-tuy=0.$$

For $t\neq 0$ this is the graph of $[u:v]\mapsto[tu:v]$. At $t=0$, its equation is $xv=0$, so the fiber is the union of $\mathbb{P}^1\times\{[0:1]\}$ and $\{[1:0]\}\times\mathbb{P}^1$. It is not a graph: the first projection has an entire projective line over $[1:0]$.

The family is flat. The four standard affine charts have equations $x-tu$, $1-tuy$, $xv-t$, and $v-ty$, respectively. Each is a primitive polynomial over $k[t]$, so its quotient ring is $k[t]$-torsion free by Gauss's lemma. A torsion-free module over the principal ideal domain $k[t]$ is flat. All fibers are divisors of bidegree $(1,1)$, with Hilbert polynomial $2m+1$ for $\mathcal{O}(1,1)$. The Hilbert scheme retains the limit, while its open morphism locus omits it.

References. The explicit polynomial parameterization is recalled in [1], §2.1B, printed pp. 77–78. Formula (1.6.1) agrees with its graph construction in §2.1C.

1.6B Maps, embedded curves, and parameterizations

Let Y be a projective variety. A point of $\mathrm{Mor}(\mathbb{P}^1, Y)$ contains the map from a fixed projective line. A point of $\mathrm{Hilb}(Y)$ contains only an embedded subscheme of Y. These are different moduli problems.

Example 1.6.4 (Lines in projective space)**.** The lines in $\mathbb{P}^N$ are parameterized by $\mathrm{Gr}(2, k^{N+1})$, where the Grassmannian now parameterizes two-dimensional *subspaces*. Its universal line is

$$\{(p,\ell)\in\mathbb{P}^N\times\mathrm{Gr}(2,k^{N+1}) \mid p\in\ell\}.$$

The Grassmannian has dimension $2(N-1)$. The degree-one morphism space has dimension $2(N+1)-1=2N+1$, larger by $3=\dim\mathrm{PGL}_2$. An embedded line has a three-dimensional family of parameterizations. One can recover the space of lines by the geometric quotient of this embedding locus by reparameterization.

Example 1.6.5 (A multiple cover is not an embedded conic)**.** The map $\mathbb{P}^1\to\mathbb{P}^1$ given by $[u:v]\mapsto[u^2:v^2]$ has degree two. Its graph has polynomial $3m+1$ for $\mathcal{O}(1,1)$. Its scheme-theoretic image is just $\mathbb{P}^1$, with polynomial $m+1$ for $\mathcal{O}(1)$. Its pushforward one-cycle is twice the line. Thus taking the image alone loses the degree, while taking the graph retains the map. This distinction remains necessary when constructing spaces of rational curves.

Remark 1.6.6 (The next deformation-theoretic step)**.** For a smooth projective source C and a smooth target Y, the familiar tangent-space formula for maps is

$$T_{\mathrm{Mor}(C,Y),[f]} \simeq H^0(C, f^*T_Y).$$

For a smooth embedded curve $C \subset Y$, the Hilbert tangent space is instead

$$T_{\mathrm{Hilb}(Y),[C]} \simeq H^0(C, N_{C/Y}), \qquad N_{C/Y} = (\mathcal{I}_C/\mathcal{I}_C^2)^\vee.$$

When $C \simeq \mathbb{P}^1$ and f is its embedding, the sequence

$$0 \longrightarrow T_C \longrightarrow f^*T_Y \longrightarrow N_{C/Y} \longrightarrow 0$$

and $H^1(\mathbb{P}^1, T_{\mathbb{P}^1}) = 0$ give

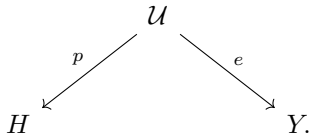
$$H^0(C, N_{C/Y}) \simeq H^0(C, f^*T_Y)/H^0(C, T_C).$$

This removes the infinitesimal reparameterizations. These formulas are stated here to connect the parameter spaces with subsequent lectures; their deformation-theoretic proofs are not part of the Hilbert existence argument.

References. For the map tangent space, see [1], Proposition 2.1.16 and Corollary 2.1.17, printed pp. 80–81. For embedded deformations and the Hilbert tangent space, see [4], Chapter I, §2, and [3], II, §8 for the conormal sequence.

1.6C Incidence as the starting point for VMRTs

Once a family of curves is parameterized by a scheme H, its universal family has two maps



Here p records which curve a point belongs to, and e evaluates that point in the target. For embedded families, $\mathcal{U} \subset Y \times H$ and e is the ambient projection. The fiber $e^{-1}(y)$ is the incidence scheme of members of the family through y, with the point on the member retained.

For lines in $\mathbb{P}^N$, the family through y is $\mathbb{P}^{N-1}$. Sending a line through y to its tangent direction identifies this space with $\mathbb{P}(T_{\mathbb{P}^N,y})$, with projective space understood here as the space of tangent lines. This is the simplest example of the geometry underlying a variety of minimal rational tangents.

For general rational-curve families, one must first choose an appropriate irreducible family and handle parameterizations, normalization, and the smoothness needed to define a tangent direction. The Hilbert and morphism schemes constructed in this lecture supply the parameter spaces and universal families on which those later constructions depend.

1.6D Exercises

Exercise 1.6.7. For a degree-d morphism from a smooth curve C of genus g to $\mathbb{P}^N$, use a source polarization of degree a to compute its graph polynomial. Explain why it is independent of the chosen parameterization when $C \simeq \mathbb{P}^1$.

Exercise 1.6.8. Compute the Hilbert polynomial of the union of two distinct intersecting lines in $\mathbb{P}^2$ by the sequence

$$0 \to \mathcal{O}_{L_1 \cup L_2} \to \mathcal{O}_{L_1} \oplus \mathcal{O}_{L_2} \to \mathcal{O}_{L_1 \cap L_2} \to 0.$$

Compare it with a smooth conic and with two skew lines in $\mathbb{P}^3$.

Exercise 1.6.9. Let $T = T_1 \amalg T_2$ be a disjoint union of two nonempty schemes. Give a Hilbert family whose polynomial is 1 on T_1 and 2 on T_2. Describe its classifying map into $\mathrm{Hilb}(X/S)$ and explain why it does not belong to a single fixed-polynomial functor evaluated on T.

Hints and checks. For the first exercise, the line bundle on the graph has degree $a + d$, so Riemann–Roch gives $(a + d)m + 1 - g$. Reparameterization of $\mathbb{P}^1$ does not change either degree, and $g = 0$.

For the second, the intersection point contributes 1, giving $2(m + 1) - 1 = 2m + 1$. Two skew lines are disjoint, so their polynomial is $2m + 2$. The different constant terms record different arithmetic genera.

For the third, one possible choice is $X = \mathbb{P}^1_k$, $S = \operatorname{Spec} k$, and $T_1 = T_2 = \operatorname{Spec} k$. Take one fixed reduced point over T_1 and two fixed distinct reduced points over T_2. The classifying map sends the two components into $\mathrm{Hilb}^1(\mathbb{P}^1)$ and $\mathrm{Hilb}^2(\mathbb{P}^1)$, respectively. The resulting family is flat over the disjoint union, but its polynomial is not a single constant function on that union.

References and Reading Guide

For a first reading, [3] supplies the cohomological facts, [5] supplies the construction of Hilbert schemes, and [4] connects the parameter spaces to rational curves. The following locators indicate where to continue a proof omitted from this lecture.

Topic	Reading location
Flatness and exactness on fibers	[3], III, §9; the Tor calculation in §1.2A above.
Hilbert polynomial, dimension, degree	[3], I, §7; [6], Tag 08AC, Tag 08AD.
High twists and pushforward	[3], III, §8 and §12, Theorem 12.11; [6], Tag 0D4E.
Flat base change for cohomology	[3], III, Proposition 9.3.
Regularity and boundedness	[5], §2; [6], Tag 08AG.
Flattening stratification	[5], §4, Theorem 4.3.
Projective Hilbert scheme theorem	[5], §5, Theorem 5.1; [4], I, §1.
Open isomorphism condition	[6], Tag 0G0Q; [5], Theorem 6.5.
Morphism schemes from graphs	[5], Theorem 6.6; [1], Theorem 2.1.11, pp. 78–79.
Tangents and obstructions for curves	[4], I, §2; [1], §2.1C–D, pp. 80–85.

Bibliography

[1] *Birational Geometry Seminar: Revised Chapters 1–5 and Appendices*. Unpublished seminar notes supplied by the author. Chapter 2, especially §2.1B–D, printed pp. 77–85. The page numbers in this lecture refer to the printed pagination, not the PDF viewer's page count.

[2] A. Grothendieck, *Techniques de construction et théorèmes d'existence en géométrie algébrique IV: les schémas de Hilbert*, Séminaire Bourbaki, 1960/61, Exposé 221, pp. 249–276. Original article and linked erratum at Numdam.

[3] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977. Publisher and DOI. Chapters I–III supply the projective, flatness, and cohomological background; IV, §1 supplies Riemann–Roch for curves.

[4] J. Kollár, *Rational Curves on Algebraic Varieties*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, 32, Springer, 1996. Publisher and DOI. Chapter I, §1 for parameter spaces and §2 for deformation theory; Chapter II for rational curves.

[5] N. Nitsure, *Construction of Hilbert and Quot Schemes*, 2005, arXiv:math/0504590. Published in *Fundamental Algebraic Geometry: Grothendieck's FGA Explained*, Mathematical Surveys and Monographs 123, American Mathematical Society, 2005, pp. 105–137. Section numbers are those of the 36-page arXiv version; page locators in the text also refer to that version.

[6] The Stacks Project Authors, *The Stacks Project*, stacks.math.columbia.edu. References use stable tags rather than chapter or lemma numbers, which may change. Relevant tags: Tag 051D, Tag 08AC, Tag 08AD, Tag 08AG, Tag 0D4E, Tag 02K9, Tag 0B94, Tag 0CZX, Tag 0G0Q, and Tag 023R.